EMERGENT RATIONALITY IN LARGE LANGUAGE MODELS

GREGOR BETZ (DEBATELAB & COMPPHIL²MMAE, KIT)

¶1. This is work in progress. It is emerging from a continuous and deep conversation Christian Seidel and I have been engaged in for several years.

I. Thinking Clearly about Competence with E. Sosa (Sosa, 2010)

¶2. Skills, Shapes and Situations. An agent is competent to the extent that they have the right constitution (possess the relevant *skill*), are in good condition (are in proper *shape*), and face favorable external circumstances (are positioned in a suitable *situation*). A competent tennis player, able to win the next match, has good serving skills; is fit, healthy and sober; and does not face a biased judge.

¶3. Accurate, Adroit and Apt Performances. With respect to a given competence, a performance is *apt* iff it is *accurate* (successful) because it is *adroit* (competently performed). Suppose the tennis player won the game (her performance was accurate). She may have done so *because* she played so well, in which case her performance was adroit (a display of her skill) and hence apt. She may have equally won with a lousy display (the performance was not adroit) because the opponent was ill, in which case the performance was not apt, although accurate.

¶4. Assessing Competence via Trigger Condition Tests. We assess an agent's skills by observing the accuracy of her acts in situations where we have reason to assume that the inner condition and external circumstances are right for the agent (provided she *is* skilled) to perform succesfully.

Inaccurate performances may be explained with an agent's lack of skill, her being ill-shaped, or her facing an unsuitable situation.

Trigger condition tests yield *non-deductive empirical evidence* that may confirm or disconfirm the presence of a skill. Accordingly, being among the top-ten tennis players worldwide for 5 years without interruption provides much stronger evidence for outstandinging skills than having beaten the no-1 player once.

Date: 2025-10-21.

II. Rationality – Plain Ryleian Starting Points

¶5. Intelligent behavior versus intellectual work. I am following G. Ryle in distinguishing intelligent conduct, like the tennis-player's smart display or the wild dogs' clever hunt, from intellectual work, as a "separate, self-moving and self-piloting activity of reflection," (Ryle, 2009/1962, p. 437). Rational thinking, as I understand it here, is of the latter kind.

¶6. Rational thinking versus freely associating. The rational thinker is engaging in intellectual work, but she is not (just) freely associating, creatively imagining, or dreaming. Rather, her thinking "has standards of its own, such that in retrospect the thinker may find fault with the way in which [she] had thought." (441)

¶7. Learning rational abilities. Rational skills are constitutive for the ability to think autonomously in line with standards of rational thought. Compliance with rational norms is learned, rational skills are acquired. For instance, "students are being deliberately trained and stimulated to think like good mathematicians or good historians or good philosophers or good composers of Latin verse or good grammarians" (441)

¶8. Universal rational skills. The norms of rationality are manifold, and so are rational skills. Some of these rational abilities are however domain-independent – these are universal thinking skills one has to master no matter whether one seeks to become a good mathematician, a good composer of Latin verse, or a good historian. Our discussion of the relation between language and thought (¶42 ff.) focuses on such universal rational skills.

III. Linguistic Understanding

¶9. Linguistically performed skills versus language skills. A linguistic performance consists in producing written or spoken utterances in a language. Many skills may be performed linguistically, i.e. exercised through linguistic performances (e.g., the skill to properly instruct a novice baker). Some skills are inevitably performed linguistically, like the judge's ability to justify their verdict, the moderator's skill to adequately summarize a debater's key point, a scholar's ability to describe a painting, or the comedian's ability to tell a joke.

From such linguistically performed skills I want to distinguish *language skills* proper. Language skills are those skills all (actual and possible) agents who fully master—can competently speak and do understand—a language have in common. For example: the ability to resolve anaphora when being prompted to do so, or the ability to turn statements into questions that inquire whether the statement is true. My hunch is that all language skills are necessarily linguistically performed (consider brain-in-a-vat thought experiments), but not every necessarily linguistically performed competence is a language skill (recall joking from above). Moreover, performing a skill linguistically typically requires one to master language skills.

In this picture, linguistic understanding is not a particular language skill, but the whole cluster of all such skills. Let us try to characterize this cluster of language skills.

¶10. **Linguistically situated skills.** Skills are exercised in *situations* by an agent in a given shape (¶2). If every situation in which some skill may possibly be exercised can be fully provided as text, then we say that the skill is *linguistically situated*.

Telling a joke is necessarily linguistically performed, but not linguistically situated. Reacting adequately to a joke (*not* to its performance) is linguistically situated, but not necessarily linguistically performed.

¶11. **Wittgensteinian language skills.** Let's say that a language skill is Wittgensteinian iff it consists in the ability to properly and correctly use language in a specific way (e.g. with regard to specific norms, in specific contexts, or in order to reach specific goals). Wittgensteinian language skills are linguistically performed *and* linguistically situated. Such Wittgensteinian language skills are legion, and can be individuated at different scales.

∼∼∼

For example, the *Common European Framework of Reference for Languages (CEFR)*[1] distinguishes 90 language skills, from "Acting as an intermediary in informal situations (with friends and colleagues)" to "Watching TV, film and video". CEFR distinguishes, for each skill, up to 10 competence levels defined by multiple descriptors. For the language skill "monitor and repair", CEFR describes competence levels (ranging from highest to lowest) as:

- Can backtrack and restructure around a difficulty so smoothly that the interlocutor is hardly aware of it.
- Can backtrack when they encounter a difficulty and reformulate what they want to say without fully interrupting the flow of language.
- Can self-correct with a high degree of effectiveness.
- Can often retrospectively self-correct their occasional "slips" or non-systematic errors and minor flaws in sentence structure.
- Can correct slips and errors that they become conscious of, or that have led to misunderstandings.
- Can make a note of their recurring mistakes and consciously monitor for them.
- Can correct mix-ups with the marking of time or expressions that lead to misunderstandings, provided the interlocutor indicates there is a problem.
- Can ask for confirmation that a form used is correct.
- Can start again using a different tactic when communication breaks down.

Observation: Some of these descriptors refer to mental activity (like conscious monitoring), or presuppose extra-linguistic abilities (gesturing, pointing to an object) and context (a video display). I suppose many descriptors can in fact be rephrased so as to involve pure linguistic performance. Even if we drop those which can't, a large variety of Wittgensteinian language skills as defined here remain.

¶12. **Intelligent language use and linguistic reflection.** Ryle's distinction between intelligent behavior and intellectual reflection extends to linguistic performances, too. Starting with Ryle's example, we may contrast:

(i) The tennis player has been playing very intelligently, allowing her to win with minimum effort.	(i') The tennis player is carefully devising a match plan.
(ii) I am typing an email informing a candidate they have failed the exam.	(ii') I am pondering how to write an email to inform candidates who have not passed.
(iii) I am backtracking on my previous remark, correcting the mistake I made.	(iii') I am wondering whether I just made a mistake, and am contemplating different ways to potentially flag and correct it.

Linguistic reflection is a special kind of intellectual reflection, a "separate, self-moving and self-piloting activity" (see ¶5).

Linguistic reflection is linguistically performed. One cannot think about intelligent language use without using language. (This also holds for most, but, as Ryle usually insists, not necessarily for *all* kinds of reflection. A counterexample that convinces me: I may reflect on how to picture-prove a geometric theorem by sketching, exploring, manipulating, and comparing geometric drawings.)

A corollary of the above thought: Linguistic reflection is itself a form of intelligent language use; this particular intelligent language use is characterized and differentiated from other intelligent language use by its specific domain and goals (e.g., preparing, exploring, planning language use rather than reflecting on, say, tennis playing), its normative standards (grammatical correctness, stylistic coherence, semantic consistency, rather than ITF rules or shot quality), and the particular vocabulary that comes along with these. Since the domain of linguistic reflection are linguistic performances (linguistic reflection refers to linguistic performances), it may be described as 2nd- / higher-order intelligent language use.

An act of linguistic reflection is accurate iff the linguistic performance which is the object of the reflection is apt (or, at least, adroit, depending on whether the reflection did or did not cover alternative contingent conditions and circumstances). Likewise, the tennis player's match planning was successful in case she won due to her impeccable performance (or performed heroically but lost nonetheless because of unpredictable interferences).

Is the ability for linguistic reflection a language skill? To think about this clearly, the following distinction seems helpful: Every linguistic performance is *eo ipso* a (more or less accurate) display of language skills. But many linguistic performances are much more than that, namely exercises of various other skills (some of which are inevitably linguisticlly performed, like telling a joke, while others aren't, like providing consolation). That means one may reflect on a linguistic performance "from different angles," depending on which is the particular skill one takes the performance at hand to be a display of. With this in mind we may now answer our question:

The ability for linguistic reflection of performances *of language skills* is itself a language skill, because competent speakers who fully master a language also master the vocabulary and the norms that allow one to refer to, and to evaluate linguistic performances as exercises of language skills. Moreover, and more generally, the ability to produce meaningful (not necessarily successful or apt) linguistic reflection is a language skill, too, because producing meaningful reflection is. However, the ability to ponder about how best to tell a joke is *not just* a language skill.

¶13. Linguistic understanding as competent language use. I now want to give force to the idea that every language skill is Wittgensteinian.

This idea casts linguistic understanding as competent language use and insists that all it takes to be a competent speaker of a language is to master its vocabulary and norms, i.e., to be able to make intelligent use of that language in diverse contexts. Such a competent speaker can be said to understand a language very much like a chess player understands chess to the extent that she (the player) can make intelligent use of her pieces.

This seems very much a Quinean view (Quine, 1960, 1992).

(None of all this entails that a competent speaker, who understands *a language*, understands *the claims set forth*, or *the subject matter covered* in an arbitrary text written in that language—which would, in particular, require the ability to correctly answer all questions of understanding re that text. This is because even a competent speaker might, for example, fail to follow a highly complex argument sketched in a, say, legal text, or might lack the required empirical background knowledge to correctly interpret a text published in a scientific journal.)

An argument it would be nice to have:

(1) Premise. A language skill is a skill shared by all (actual and possible) competent speakers of a language.
(2) The conjunction of all necessary conditions for being a competent speaker of a language is sufficient for understanding a language.
(3) Every language skill shared by all competent speakers of a language is Wittgensteinian.
(4) *This seems to support*: Linguistic understanding consists in the ability to competently use a language.

The idea that every language skill is Wittgensteinian demarcates a deflationary notion of linguistic understanding from:

- *Representationalism*: Linguistic understanding requires (conceptually) that one forms mental representations corresponding to the propositions expressed in that language, possibly presupposing that one stands in suitable causal connection to the entities and events referenced in that language.
- *Phenomenological accounts of understanding*: Linguistic understanding requires (conceptually) conscious cognition.

IV. Abstract Language Models as Representations of Normative Regimes

¶14. Rule-governed practices. A *rule-governed practice* comprises activities that can be assessed according to specific evaluative standards (rules, criteria, overarching goals) and are typically performed by agents who, in doing so, are guided by these standards. Tennis is a rule-governed practice. Debating is, too. And so is stand-up comedy.

¶15. Normative regimes. The cluster of norms, rules, criteria or evaluative standards which govern a practice constitutes the practice's *normative regime*. The normative regime of tennis, for example, includes the official rules of the game, dresscode, fairplay principles, game tactics, general athletic criteria, as well as techniques and methods for performing specific tennis moves.

¶16. Codices. I conceive of a normative regime as an abstract entity which may explicated in a *codex*. One and the same regime can be explicated (partially or fully, approximately or precisely) in different codices. Much like different statements can articulate one and the same proposition. The official ITF Rules of Tennis, for example, are a codex that partially explicates the normative regime governing tennis. Likewise, a debater's handbook is an incomplete articulation of the regime governing debating. Or, an ethical theory is a codex that explicates the normative regime governing our moral practice.

Normative regimes can be explicated and articulated following different *schemes*, e.g., involving rules and norms, goals and axiological standards, skills and virtues, or a mixture of all.

Frequently it's unclear and controversial how best to explicate a normative regime. And one reason for this being so is that, in many cases, there is no simple algorithm (list of deterministic rules) that adequately, fully and precisely describes the normative regime.

¶17. Abstract Language Models. An *abstract language model* is a mathemtical structure defined over a given finite vocabulary (e.g., words, characters, or tokens) that specifies for every possible sequence of input tokens the next-token-probabilities. As such, a language model can be thought of as a *complete* list of probabilistic rules which prescribe what to "do" next come what may. Unlike in a classic algorithm, the language model's rules are however not deterministic, but probabilistic.

Qua being complete, a language model not only provides next-token probabilities, but encodes in fact the conditional probability of any sequence of output tokens given any input sequence. A language model tells, for *any* input, how likely an *arbitrary* output is.

¶18. Abstract language models as codex schemes for explicating normative regimes. Being complete and probabilistic turns abstract language models into powerful theoretical devices for explicating normative regimes.

This normative interpretation of abstract language models seems under-appreciated and runs against the way language models are typically pitched in textbooks or critically discussed by scholars—e.g., as fuzzy syntax (Russell & Norvig, 2016) or descriptive corpus statistics (Norvig, 2017).

¶19. A representation theorem for normative regimes (but no proof), and a new perspective on LLMs. I posit:

1. Every normative regime (governing a linguistic practice) can be represented by a language model.
2. Two normatives regimes (governing a linguistic practice) which can be represented by one and the same language model are identical.
3. Sampling from a language model that represents a normative regime yields a practice that is governed by the corresponding regime.

Here, a language model *represents* a normative regime if it correctly encodes, for arbitrary initial conditions (input sequences) and every possible reaction (output sequences) the probability that, according to the normative regime, the reaction is adequate given the initial situation. These probabilities also encode the degree to which a reaction is required (probability close to 1) or forbidden (probability close to 0) according to the normative regime.

This perspective allows us to understand LLMs as built machine learning systems that seek to approximate the abstract language model which encodes the normative regime governing our linguistic practices.

∼∼∼

An alternative to contemplate here is whether a normative regime might also be represented through an envelope, or an ensemble of abstract language models.

V. Varieties of Emergence

¶20. Scaling emergence. Large Language Models (LLMs) have been observed to acquire new capabilities through scaling—by increasing either model size or pre-training data. This phenomenon, where larger models exhibit skills that their smaller counterparts lack, has become a consistent finding across multiple studies and model families.

The empirical evidence for such emergent skills is overwhelming (Berti, Giorgi, & Kasneci, 2025; Wei, Tay, et al., 2022), and covers diverse abilities such as: in-context learning abilities (Brown et al., 2020), acquisition of world knowledge (Petroni et al., 2019) and commonsense (Da et al., 2021), chain-of-thought reasoning (Wei, Wang, et al., 2022), the capacity for effective self-critique (Kamoi et al., 2024), and moral self-evaluation capabilities (Bai et al., 2022). While some capabilities like instruction following require explicit fine-tuning training (Ouyang et al., 2022), many others appear with increased scale.

¶21. Curricular emergence. Sometimes a specific skill is said to be emergent in an LLM if the model has acquired the skill without having been specifically trained on that skill. This would be the case if the training data (typically a large text corpus) doesn't contain any instances of successful skill applications, evaluation metrics or reward functions used during training do not measure proficiency in that skill, and neither model architecture nor training hyper-parameters have been chosen to equip the model with the particular skill in question.

Given the ridiculous size of today's AI systems and their training data, it has been notoriously difficult to empirically verify these conditions—though Betz, Voigt, & Richardson (2021) attempt to demonstrate, in controlled experiments, the curricular emergence of complex deductive inference skills. The question whether modern LLMs do possess curricular-emergent skills is highly contested (see ¶25 ff.).

¶22. Information-theoretic emergence. Information-theoretic emergence describes the epistemic / inferential relations between a micro-level and a macro-level description of a (complex) system. Macro-level properties $\mathbf{y}$ are said to be information-theoretically emergent relative to micro-level properties $\mathbf{x}$ in system S iff the only way to infer (know) that S has properties $\mathbf{y}$ given the assumption it has properties $\mathbf{x}$ is to fully simulate the microdynamics of S (Bedau, 1997).

On a micro-level, LLMs can be described as neural next-word prediction machines that, having been trained on a large text corpus, represent the joint probability distribution of a language's token sequences. On a macro-level, the behavior of these systems can be described by normatively (logico-semantically) characterizing the linguistic output produced by these systems given meaningful natural-language queries. Arora & Goyal (2023) judge, and I agree, that there is no easy route to infer such semantic macro-level properties from the stochastic micro-level description.

This may also explain why the performance of neural LLMs has surprised most ML engineers, AI experts, cognitive scientists, and NLP scholars. Language models are not only fairly simple, but indeed a relatively old technology—which was originally conceived by Markov more than 100 years ago (Markov, 2006) and has been comprehensively studied henceforth, e.g. by Shannon (1948) and Chomsky (1969). So the stochastic micro-level properties of LLMs have been fully understood for decades, but this didn't suffice to see which semantic macro-level properties these systems have once being scaled up, only simulations have been (and are) revealing these.

¶23. Ontological emergence. The concepts of weak and strong emergence have been engineered in philosophy of mind and philosophy of science (O'Connor, 2021) to characterize the metaphysical (= conceptual) relationship between two apparently different kinds of things. For example, regarding philosophical theories about the relation between mind and body, emergence positions reject both dualism (mental and physical states are two different, conceptually independent properties) and reductionism (mental states are nothing but physical states), and posit, as intermediary views, varying degrees of interdependence between mental and physical states.

We may adapt this traditional classification of positions in philosophy of mind in order to characterize potential stances regarding the fundamental relation between language and thought, or, more precisely, between language skills (see ¶9) and rational competences.

dualism: Language skills on the one hand and rational skills on the other hand are conceptually and metaphysically entirely independent, even though the degree to which such skills are realized and dynamically evolve might contingently depend on each other.

strong emergence: Rational skills modally supervene on language skills, while otherwise being largely independent of the latter; they are, e.g., not functionally realized through language skills.

weak emergence: Rational skills are conceptually and metaphysically distinct from language skills, but strongly depend on the latter, for example by functional realization.

reductionism: Rational skills are nothing but, and ultimately identical with language skills.

As we will see, it is fruitful to think about LLMs in terms of these alternative propositions.

¶24. Wanted: A unified account. This work aims at developing a systematic and unified understanding of emergent rationality in LLMs that integrates the four dimensions of emergence above.

VI. Genuine Generalization versus Stochastic Parrots

¶25. The controversy. The widespread observation of scaling-emergence has sparked an ongoing debate about how to interpret modern LLMs (Elangovan, He, & Verspoor, 2021; Wang et al., 2025), which can be traced back to fundamental methodological differences (Norvig, 2017): Are LLMs genuinely generalizing or merely sophisticated pattern matchers?

¶26. The Genuine Generalization Argument. On one account—which we may dub the "genuine generalization view"—, scaling-emergence *is evidence for* curricular-emergence. So suppose skill s emerges when scaling model $\mathcal{M}^{\ominus}$ to model $\mathcal{M}$. The gist of the argument, which is rarely stated explicitly, is that model $\mathcal{M}$, while exhibiting s, has not been trained on that skill because, otherwise, the smaller model $\mathcal{M}^{\ominus}$ would have been trained on, and hence possess that skill, too. Argument A:

(1) $\mathcal{M}$ exhibits skill s but there is a smaller $\mathcal{M}^{\ominus}$ similarly and successfully trained that does not.
(2) *Assumption for the sake of the argument:* $\mathcal{M}$ has been specifically trained on s.
(3) If $\mathcal{M}$ has been specifically trained on s, and $\mathcal{M}^{\ominus}$ has been similarly and successfully trained, then $\mathcal{M}^{\ominus}$ has been trained on s.
(4) If a model has been trained on s, and training has been successful, then it possesses skill s.
(5) *Thus:* $\mathcal{M}^{\ominus}$ exhibits skill s.
(6) *Thus (by reductio):* $\mathcal{M}$ has not been specifically trained on s.
(7) *Thus (by definition):* Skill s is curricular-emergent in $\mathcal{M}$.

¶27. The Argument from Imitation. A contrarian account—the "stochastic parrots view" (Bender et al., 2021)—has it that LLMs simply pick up statistical patterns they see during pretraining, which gives rise to an *Argument from Imitation* against curricular-emergence. Argument B:

(1) $\mathcal{M}$ has skill s.
(2) If an LLM has skill s, it has seen sufficiently many apt performances of s in training.
(3) If an LLM has seen sufficiently many apt performances of s during training, then the training corpus contains sufficiently many (accordingly labeled) examples of apt performances of s.
(4) *Thus:* The training corpus of $\mathcal{M}$ contains sufficiently many examples of apt performances of s.
(5) If an LLM's training corpus contains sufficiently many outstanding instances of apt s-performances, then the LLM was trained on skill s.
(6) *Thus (by definition):* Skill s is not curricular-emergent in $\mathcal{M}$.

¶28. Dialectics: Balance of Arguments? The arguments A and B lead to contradictory conclusions: Accepting one argument compels one to deny at least one premise in the other. Proponents of the "stochastic parrots view" may, for example, question premise (A.4) and posit that smaller models simply fail to pick up complex patterns present in a training corpus, which are however recognized and learned by larger models. Proponents of the "genuine generalization view," in contrast, may refute key premise (B.2) in the Argument from Imitation. This is precisely what the theory of emergence proposed by Arora & Goyal (2023) suggests.

VII. Domains and Hierarchies of Skills

¶29. Analytic (apriori) versus empirical skills. The ability to tell whether a natural number is even doesn't seem to depend on what the world is actually like, whereas the skill to name the capital of every country in the world heavily "involves" empirical and contingent knowledge. These examples hint at a general distinction between *analytic* and *empirical* skills, which we may capture in terms of counterfactual robustness or *possible-worlds-invariance*: A skill s is analytic iff an agent who possesses s in this world does so in every other conceptually possible world; otherwise skill s is empirical.

That a skill s is empirical means that for any agent who possesses skill s in this world—i.e., passess suitable trigger-condition tests if in good shape—there's another conceptually possible world where this very same agent wouldn't pass most such tests, even if the agent were in good shape and the situation were right.

∼∼∼

In defining analytic skills (that may be acquired apriori) with respect to all *conceptually* possible worlds, I intend to exclude possible worlds with radically different laws of nature, reflected in fundamentally different conceptual frameworks, from the counterfactual robustness test.

¶30. **Language skills are analytic.** An agent who masters, say, Swedish in *this* world does so in every other conceptually possible world. She may, in that other possible world, not be in good *shape* to actually speak, or the *situation* might typically not be conducive for her to understand what is being said, but this does not prevent her from having the language *skill* to speak and understand Swedish.

While every language skill is analytic, many necessarily linguistically performed skills are not. Your jokes could knock over people today, but might suck in an alien future, no matter how hard you try.

¶31. **Basic and complex skills.** Skills can sometimes be decomposed into more basic component skills: The tennis player's competence to serve, e.g., comprises the skill to throw, to swing, and to strike. To serve competently, it takes the tennis player however more than to be competent in each of the basic skills. In this particular case, performances of the skills need to be approriately coordinated. More generally, we shall say that component skills can be "configured," and a complex skill is realized through a *configuration of more basic skills*. A configuration is a schema for the interdependent and coordinated execution of component skills.

Skills are basic *relative* to a complex skill. Sometimes a basic skill, which is a component in a configuration realizing a complex skill, can itself equally be conceived of as complex skill, which is realized through other component skills. So basic skills are not necessarily atomic skills. I neither affirm nor deny that there are atomic, truely fundamental skills.

∼∼∼

The notion of a basic skill may help to explain the idea of "axiomatizing" a collection of skills. Thus, a set of skills S is axiomatized by basic skills $B \subseteq S$, $\bar{B} = S$, iff every non-basic skill $s \in S \setminus B$ can be realized through a configuration of basic skills from B.

¶32. **Multiple realizability of complex skills.** Complex skills may be realized through different configurations of basic skills.

Consider for example the tables from 1 to 10. All kids in the group perfectly master these. The first one competently retrieves the correct result for any task from the tables she has memorized. The second one, building on their skill to count and sum, consecutively adds the multiplicand to arrive at the correct result. The third uses distributivity to split the calculation into multiplications that can be solved with the tables from 1 to 5. The fourth one …

∼∼∼

For skill $s \in S$ and configuration $\sigma \in \Sigma$ ($\Sigma \subset \bigcup_i S_1 \times ... \times S_i$), we write $\pi(\sigma) = s$ to express that σ realizes s. Multiple realizability amounts to the claim that $\pi(\sigma_1) = \pi(\sigma_2)$ does not necessarily imply $\sigma_1 = \sigma_2$.

¶33. **Approximately realizing complex skills through retrieval and rule following.** Consider a skill that concerns mastery of a rule-governed practice. (We don't need to assume that every skill is of this kind.) One particular way to realize this skill is to recall the rule(s) and to apply them to the situation at hand. Let's call this the *retrieval-and-rule-following* configuration scheme.

The component skills of any retrieval-and-rule-following configuration are: the skills to retrieve and state the rules relevant for the situation at hand, and the skills to apply and follow the rules. The retrieval skill relates to the system at hand having *reliable memory* for storing and recalling texts—which we shall assume henceforth. The rule-following skill may be conceived of as *primitive syllogistic skills* which consist in drawing simple FOL inferences, and are hence basic language skills.

For illustration, let's return to our math youngsters. The fourth kid has carefully memorized a comprehensive method (involving multiple "tricks") for solving the tables. First thing she did when starting the test was to recall and write down the exact general algorithm for solving any tables task. She then simply applied this algorithm to every task in the test, following the algorithm's rules step by step.

Some norm-governed practices may not be fully explicateble in terms of an algorithm (e.g., as Kuhn famously claimed, scientific theory choice or paradigm change). In this case, what a *retrieval-and-rule-following* configuration may nonethelss yield is a (good enough? arbitrarily precise?) approximation of the complex skill at hand.

¶34. **Appropriately configuring skills with workflows and planning as meta skill.** Configuring basic skills into a scheme such that the interdependent execution of these skills realizes a more complex one is itself a skill—namely a higher-order, or meta ability.

Assume the second kid from above, a few weeks ago, had forgotten to learn for the math test and figured out for herself, during the test, that she can solve multiplication tasks by simply adding up one of the coefficients for a given number of times. She has thus exercised the meta-skill of configuring more basic skills into a complex one, and now possesses the complex skill (which is realized through a specific configuration).

What exactly is this meta skill? One obvious way to conceive of it is as a particular kind of intellectual reflection (¶12) that involves the agent pondering and planning how to (better) combine basic skills into a workflow that allows the agent to achieve specific goals, thusly improving her competence in a complex skill.

However, we don't need to maintain that such *System-2* planning is the only way to exercise skill-configuration competence; intelligent agents may equally be able to effectively combine (e.g., after a few trials) basic skills into an effective configuration so as to realize a new complex skill.

If the basic component skills as well as the complex skill are linguistically situated and performed, the meta-skill to effectively configure the basic skills so as to realize the complex one—no matter whether this involves explicit planning and intellectual work or not—is itself linguistically situated and performed.

VIII. Universalilty of Cloze Tests

¶35. Cloze test solving skills. Cloze tests ask testees to fill in missing words or phrases. Every text can be turned into a cloze test (indeed many). Snippet A,

> Does Sarah have family? Sarah is Pete's sister, so Pete is her __________.
> Both are living in Paris, the capital of __________.

Likewise, multiple choice tests or Q/A quizzes can be cast as cloze tests, too, where testees are required to replace a suitably placed mask with their answer.

As long as there's a single correct substitution for each placeholder, the ability to solve cloze tests (of a particular kind) simply amounts to the ability to provide the correct answers.

However, sometimes more than one substitution is correct, like for the first mask above, where "brother" seems most obvious, but both "sibling" or "relative" are also viable. I will assume, and maybe that is not an innocent assumption, that whenever there is more than one legal substitution in a cloze test, there is a probabilistic degree to which a given substitution is objectively adequate. For instance, "brother" might be the correct answer with probability 0.8, while "sibling" and "relative" are correct with probability 0.1 each. Now, a testee fully masters a class of cloze tests iff her disposition to give a certain answer matches the objective probability of that answer being correct. Assuming that dispositions imply propensities (likely not innocent either), meaning that behaviour is governed by conditional probabilities, a testee's level of competence can be conceived of as the agreement between her own probabilistic disposition and the objective normative probability distribution over possible answers.

∼∼∼

Let $p(\cdot|z_i)$ be the objective normative probability distribution over possible answers to cloze tests $z_1, z_2, \ldots$, and $q(\cdot|z_i)$ be the testee's propensity. Then the testee's level of skill can be construed as the KL divergence between p and q: $D(p|q) = \sum_x p(x|z_i) \log \frac{p(x|z_i)}{q(x|z_i)}$.

¶36. Analytic and empirical cloze tests. Consider the two placeholders in Snippet A above. Accurately replacing the first placeholder requires only conceptual understanding, the skill to do so is therefore counterfactually robust in the sense of **¶29** and the corresponding cloze test solving skill is analytic. This is not true for the second slot, which "tests geographic knowledge" (we'd informally say) and hence requires empirical skill.

More generally, we may characterize an entire class of cloze tests in terms of whether the skill to accurately solve all of them is analytic or empirical.

¶37. Wittgensteinian skills are cloze test solving skills. I assume:

- For every skill s which is both linguistically performed and linguistically situated, there is a class of cloze tests Z such the an agent's level of s-competence *de facto* equals her competence to solve Z-tests [conditional on the agent possessing all other competences required for solving each Z-test].

The assumption says that we can track lingusitically performed and situated skills by cloze tests. This appears highly plausible given the extreme versatility of cloze tests. Yet, moreover and more far reaching, I posit a stronger constitutional thesis:

- For every Wittgensteinian language skill l there exists a class of cloze tests Z such that skill l *consists in* the [conditional] ability to solve Z-tests [assuming all other skills required for solving each Z-test can be exercised aptly].

Language skills ultimately *are* abilities to solve cloze tests. That almost seems to follow from what has been said above (esp. **¶13**): Competent language use is the ability, when speaking and writing, to adequately decide what to say or text next. But that, in fact, boils down to nothing but solving cloze tests of a particular type, which resonates with Quine:

> In this matter of understanding language, there is thus a subtle interplay between word and sentence. In one way the sentence is fundamental: understanding a word consists in knowing how to use it in sentences and how to react to such sentences. Yet if we would *test* someone's understanding of a sentence, we do best to focus on a word, ringing changes on its sentential contexts. (Quine, 1992, p. 58, *my emphasis*)

An additional justification for the constitutional thesis stems from conceiving abstract language models as codices that may comprehensively encode the normative regime governing our linguistic practices (**¶18**). This is because an abstract language model can be understood as a huge collection of probabilistic cloze-tests. An agent that perfectly masters all cloze tests (next-word-prediction tasks) which are implied by a language model adequately representing a normative regime, *eo ipso* fully masters our corresponding linguistic practices.

In consequence, linguistic understanding as competent language use is a cluster of cloze test solving skills.

∼∼∼

Let $\mathbb{L}$ refer to the infinite set of all meaningful finite texts ($\mathcal{L}$-texts) that can be generated within a language $\mathcal{L}$. A cloze test z is tuple $\langle t, m, p \rangle$ consisting in an $\mathcal{L}$-text t, a mask m, and a multi-variate probability distribution p describing the correct substitutions for slots in m.

A particular language skill l can be individuated through the (unique) set of all cloze tests Z_l that require l-competence; solving a test $z \in Z_l$ means to exercise skill l (possibly together with further required skills).

We can hence empirically assess the l-performance of a testee by letting her solve cloze tests sampled from Z_l, while paying attention the testee is in proper *shape*, i.e., possesses any other skills beyond l that may be required to solve the tests.

IX. Learning Linguistic Skills

¶38. Setting the scene: The ontology of skills and texts (Arora & Goyal, 2023). The theory proposed by Arora & Goyal (2023), which I shall abbreviate as "AGE," assumes that there is a primitive and finite set of basic skills B, identifies any tuple of basic skills with a complex skill, and describes the process through which an *abstract learner* may acquire these skills.

We shall deviate from AGE in assuming that, in addition to basic skills B, there are configurations of (tuples of) basic skills $\Sigma \subset \bigcup_i S_1 \times ... \times S_i$, and complex skills C, where every complex skill can be realized through at least one configuration of basic skills (cf. **¶31** ff.). With $S = B \cup C$ we thus have $\bar{B} = S$. Note that we don't need to explicitly include higher order complex skills, which are realized through a configuration of complex (rather than basic) skills, in the skill graph. That is because such higher order skills can always be cast as (1-st order) complex skills configured with basic skills.

AGE assumes there exist, in addition to skills, *text pieces* which collectively form a corpus T. Every text piece is *associated* with a set of basic skills. (Interpreting the abstract model: The rough idea is that basic skills associated with a text have been used in generating and are required to comprehensively understand the text, see also **¶40**.)

The overall framework can thus be depicted as a *4-stack skill graph*, with text pieces T, basic skills B, configurations of basic skills Σ, and complex skills C as different layers stacked on each other.

We say that an arbitrary combination (tuple) of basic skills $\langle s_1, \dots, s_k \rangle$, $s_i \in B$, is *instantiated* in the text corpus T iff there is a text piece $t \in T$ that is associated with each individual basic skill $s_1 \dots s_k$ in the combination. Moreover, a complex skill $s \in C$ is instantiated in the text corpus T iff some configuration $\sigma \in \Sigma$ of basic skills, $\pi(\sigma) = s$, is instantiated in T.

¶39. **An abstract model of learning skills by observing text (Arora & Goyal, 2023).** In AGE, the abstract learner acquires basic skills by continously *observing* text pieces from a training split $T_T \subset T$. More specifically, the abstract learner, by observing a single text piece $t \in T_T$ during training, incrementally improves their performance in all basic skills associated with t.

Besides this procedure for learning basic skills, which is central for AGE, we need to detail how the learner may acquire complex skills. For simplicity, let us assume in a first approximation that the learner already masters the *meta skill* of effectively configuring basic skills so as to realize complex skills (**¶34**). Then, the learner *acquires* a complex skill $s \in C$ qua learning all basic skills s_i in a configuration $\sigma = \langle s_1, \dots, s_k \rangle$ which realizes skill s ($\pi(\sigma) = s$). Similarly, the learner *masters* a complex skill s as soon as there is a configuration σ realizing s ($\pi(\sigma) = s$) such that the learner masters all component skills of σ.

¶40. **LLMs are AGE learners.** AGE is a model of how LLMs learn. To see this, let us apply AGE to LLMs and interpret its abstract concepts.

To start with, we may conceive of basic skills as arbitrary linguistically situated and linguistically performed skills. It doesn't matter whether these skills are empirical or analytic; whether they are necessarily linguistically performed or not. Inevitably, though, some of these basic skills will be proper language skills (**¶9**).

In consequence, the complex skills in AGE, qua being realized through basic skills, are linguistically situated and linguistically performed, too.

The abstract text pieces in AGE can obviously be thought of as natural language texts, forming a text corpus (with training and test split). Each text exemplifys and illustrates specific (linguistically situated and performed) skills, all of which need to be mastered to fully understand the text. This is because "understanding a text" denotes, in this context, the ability to solve the various cloze tests that can be derived by masking the text in one way or another. Our discussion of the universality of cloze tests, specifically **¶37**, provides the background and lends support to this interpretation of AGE.

Equating linguistically situated and performed skills with cloze test solving competence allows us to see how LLMs (the learners) may gradually acquire skills by "observing" texts: Standard training of LLMs on texts proceeds (conceptually) by deriving cloze tests from the training text (next word prediction tasks), letting the LLM solve one cloze test after the other, and adjusting the internal weights depending on how much the LLM errs after each trial. So the LLM gradually improves its cloze test solving competence, and hence the corresponding linguistically situated and performed skills.

It is finally not implausible to assume that LLMs may gradually learn, during pretraining, to effectively configure basic skills. We noted in **¶34** that this meta skill may well consist in competent reflection and planning, which is itself a linguistically performed and situated skill, and might hence be easily acquired by "observing" training texts. The literature on LLM's so-called thinking-aloud and chain-of-thought reasoning skills backs this assumption with ample empirical evidence (Betz, Richardson, & Voigt, 2021; Wei, Wang, et al., 2022).

~~~

Let $\mathbb{L}$ refer to the infinite set of all meaningful finite texts ($\mathcal{L}$-texts) that can be generated within a language $\mathcal{L}$. A corpus of texts T ($\mathcal{L}$-corpus) is a subset of $\mathbb{L}$. Every text $t \in T$ from such an $\mathcal{L}$-corpus can be masked (in various ways). Assuming that the masked words and phrases are the uniquely correct answers gives a "corpus-cloze-test," which is a particular kind of cloze test (in the sense of **¶35**) whose objective normative probability function is a Dirac measure.

(Most individual corpus-cloze-tests inevitably underestimate the existing normative underdetermination; we may assume, however, that *all* corpus-cloze-tests derived from a text corpus *collectively* do represent the objective uncertainty about what is a legal substitution of a given placeholder. E.g., the corpus may contain many snippets with "Sarah is Pete's sister, so Pete is her brother" alongside some more snippets saying "Sarah is Pete's sister, so Pete is her sibling.")

During pretraining (cross-entropy loss), an LLM effectively practices to solve particular corpus-cloze-tests, namely ones where the testee is asked to predict, given a text sequence, the next word. The LLM's propensities are adjusted, one missing next word after the other, to align with the (presumably uniquely) correct answers.

Given the strong theoretical tie between linguistically situated and performed skills on the one hand and cloze-test solving competence on the other hand (**¶37**), it seems plausible to assume: Every next-word-prediction corpus-cloze-test corresponds to at least one, but typically many different linguistically situated and performed skills. Language skills proper will almost necessarily be amongst
~~~

these, because with every word one reads one inevitably learns something about basic language use, e.g., if only by observing that "blue" (or any other color world) is not a correct continuation of "… Both live in Paris, the capital of" (see also ¶9). That is how each training text, giving rise to many such cloze tests, is associated with different skills, which yields a nice interpretation of AGE's skill graph. By learning to solve cloze tests derived from a training text, the LLM hence improves its competence on the skills associated with the text.

If, in addition, the training corpus is sufficiently diverse so as to contain, for every basic language skill l ($l \in S_{\text{lang}} \subset S$), a large number of texts from which next-word-prediction cloze tests $z_1, z_2 \dots$ can be derived such that every z_i belongs to the class Z_l of cloze tests constitutive for language skill l (¶37), then we may plausibly assume that the LLM will gradually acquire every basic language skill during *comprehensive and successful* pretraining.

¶41. The emergence of complex skills. Now, the implications of all this are noteworthy:

1. We may expect to observe **scaling emergence** if more comprehensive training (achieved through scaling) enables a model to acquire more basic skills; as this typically allows for the realization (in novel configurations) of further complex skills which couldn't be configured before. The tensorization argument reconstructed in the Appendix is an attempt by Arora & Goyal (2023) to quantify this effect.
2. We may expect to observe **curricular emergence** because the model may well realize a complex skill $s \in C$ through configurations of learned basic skills without there being any configuration σ realizing s such that σ's component skills are (simultaneously) instantiated in any single text in the training corpus (but are instead distributed across many different texts).
3. We may expect to observe **information-theoretic emergence** because predicting the skills a model will learn by observing texts in a training corpus would require one to construct the *extended skill graph*: matching all basic skills to texts, and identifying all configurations of basic skills through which complex skills can be realized—which seems practically impossible.

Moreover

4. The key premise of the Argument from Imitation (¶27), maintaining that an LLM having acquired skill s implies that the LLM has "observed" instances of s during training, is just wrong.

This strips the stochastic parrots view of its main justification, and tips the balance of arguments in favour of the genuine generalization thesis.

X. On Language and Thought

¶42. Global and local weak emergence hypotheses. We may explicate the weak emergence view about the relation between language and thought (¶23) as:

Local weak emergence: Rational competence r is an analytic language skill.

Global weak emergence: Every rational competence is an analytic language skill.

These two propositions are aptly labelled "weak emergence" views in that they assume that some (or all) rational skills (¶5 ff.) can be functionally realized through, and hence *depend* on, configurations of basic language skills, while these rational skills are nonetheless individuated *independently* of language skills and certainly need not be functionally realized through those (multiple realizability, cf. ¶32).

Many philosophical doctrines imply, or rule out, weak emergence views, especially particular local weak emergence claims. For example, logicism (the view that mathematics is grounded in logic) and a Fregean account of logic imply various local emergence hypotheses about logical and mathemtical skills. Psychologism (the view that the laws of logic are ultimately psychological laws of belief), on the contrary, suggests that the global weak emergence hypothesis is wrong. In addition, empiricism dovetails nicely with a global weak emergence view, whereas a rationalist doctrine positing an ability of rational intuition or innate knowledge tends to contradict the global weak emergence view.

I shall unpack the local and global weak emergence hypotheses by describing the relationships between rational, analytic, and language skills within the AGE framework (Arora & Goyal, 2023) and characterizing, more specifically, the skill graph that adequately represents our normative, rule-governed practice. In line with our interpretation of AGE, we consider but linguistically performed and situated skills S.

¶43. Revisiting complex language and complex analytic skills. First some further notational conventions. Let $A \subseteq S$ ($A^\star =_{\text{def}} A \cap B$) be the set of all (basic) analytic skills; and $L \subseteq S$ ($L^\star =_{\text{def}} L \cap B$) be the set of all (basic) language skills.

Every complex language skill can be configured through basic language skills, so $\overline{L^\star} = L$. That is because a skill is a language skill only if is shared by all competent speakers of a language. Since there are (at least fictitional) competent speakers who possess only language skills, every complex language skills can be configured through basic ones.

Likewise, I posit that every complex analytic skill can be configured through basic analytic skills, so $\overline{A^\star} = A$. This requires us to revisit and refine the explication of analytic skills in ¶29: A *basic* skill $s \in B$ is analytic iff possessing s is counterfactually robust. A *complex* skill $s \in C$ is analytic iff there is a configuration $\sigma = \langle s_1, \dots, s_k \rangle$ realizing s ($\pi(\sigma) = s$) such that all component skills in σ are analytic ($\sigma \in A^k$)

In line with multiple realizability, all this allows for configurations of complex language and analytic skills that contain empirical skills; in other words: not every configuration of a language (analytic) skill contains only language (analytic) skills as components.

¶44. Analytic skills and language skills are identical. In ¶30 I suggested that every language skill is analytic, $L \subseteq A$. I now argue that, vice versa, every analytic skill is a language skill, $A \subseteq L$. It suffices to show that every basic analytic skill is a language skill, $A^\star \subseteq L$, since $\overline{L^\star} = L$. Now, if possessing a specific skill s is counterfactually robust across all conceptually possible worlds, the *best explanation* for this being so seems to be that s is a language skill. So, by inference to the best explanation, every basic analytic skill is a language skill.

In consequence, a skill is analytic if and only if it is a language skill, $L = A$.

¶45. Corollary: Math skills are language skills. From ¶44 directly follows that math skills, which are arguably analytic, are—inasmuch as they are linguistically situated and performed—language skills. Doesn't this amount to a *reductio*? Not necessarily, as one may argue, alternatively: First, linguistically situated and performed math skills can be reduced to logical skills which in turn can be reduced to language skills (logicism). Second, once linguistically situated and performed math skills are involved, the language relative to which one individuates language skills is not our natural language simpliciter, like Dutch and French, but covers the mathematical language, too, such that every competent speaker of this language has math skills. Third, math is a rule-governed practice, and math skills can hence be realized through a retrieval-and-rule-following configuration (cf. ¶33).

¶46. Rational skills are analytic. Following ¶5 ff., let $R \subseteq S$ ($R^\star =_{\text{def}} R \cap B$) be the set of (basic) rational skills which are linguistically situated and performed. If some $r \in R$ were analytic, then (with $L = A$) the local weak emergence hypothesis (¶42) re skill r would hold. So which, if any, rational skills are analytic?

It seems plausible that some rational skills, like the abilities to discern a plain contradiction, to resolve a double negation, or to draw a conclusion via a most simple syllogistic inference, are both basic and analytic skills.

To argue, more generally, that every rational skill $r \in R$ is analytic (and hence a language skill), one may resort to *retrieval-and-rule-following configurations* (¶33). Accordingly, the argument goes, a rational skill r is the ability to follow particular rules or norms of rationality. Any such skill can therefore be realized through a specific *retrieval-and-rule-following configuration* of basic (analytic) skills: namely the retrieval (and explicit statement) of the relevant norms of rationality, and the ensuing application of these norms (via syllogistic inference) to the specific case at hand. Therefore, and notwithstanding the possibility of realizing r through other configurations of skills, r is itself an analytic skill.

The catch here is that the argument presumes holders of skills to be able to *recall and state* arbitrary norms of rationality.

¶47. Objection: Cases where rational skills are seemingly not counterfactually robust. The following rational skills, though, seem to refute, as counterexamples, that $R = L$:

The tennis player's ability to plan a match. Her match planning skill has been outstanding this decade; no matter which opponent she faces, her plans have consistently compensated relative technical or physical disadvantages. However, in another conceptually possible world (e.g., the not so distant future where opponents' features and tactics are erratic and impossible to predict), her planning might be entirely ineffective, causing her to systematically fail trigger condition tests for match planning skills.

The teacher's ability to guide, stimulate and close a classroom discussion so that it conforms to epistemic and deliberative standards. She is arguably the best teacher in school, adored by both students and colleagues. Yet, in another conceptually possible world, where, say, students simply stop signalling agreement or disagreement in non-verbal ways, she might struggle to moderate a respectful students' discussion, be helpless when a debate escalates, and generally fail all trigger condition tests we use to assess didactic moderating and mediating skills.

The experienced engineer's ability to carefully ponder in speech and writing how to design a long-lasting, reliable and cost-efficient bridge. Having build hundreds of bridges, many of which have been internationally recognized, and invented a dozen new designs, she is at ease in explaining how to craft modern, sustainable bridges. But in another possible world, where, for example, materials available to build bridges have entirely different physical properties, she might fail to accurately reflect on how to construct the most simplest bridges whatsoever.

The lawyer's ability to counter and refute arguments of the prosecutor. Having extensively studied legal argumentation schemes in law school, and having practiced the analysis and critique of legal reasoning in terms of these schemes, she is one of the most skillfull and expensive lawyers in town, defending her clients most successfully. Still, in another possible world, with legal system and procedural norms of prosecution varying fundamentally from ours, her skill to counter and refute the prosecution's arguments will be gone.

The wrong reply to these presumable counterexamples is: We can hold on to the view that these skills are rational by re-describing them so that the counter-factual situations depicted fall outside the scope of permissible trigger condition tests. In other words, one suggests a different way of individuating the skills at hand. Yet, using that trick, you might in fact turn every empirical skill into an analytic one, by restricting trigger conditions to situations that only hold in the actual world. For example, the skill to predict the capital of a given country *given* today's international regime would thus turn out to be analytic, too. Our way of explicating the difference between analytic and empirical skills presupposes that skills are not individuated such that their scope of application is restricted to the actual world.

A more promising rebuttal of the presumable counterexamples stresses the multiple realizability of complex skills: What the cases show, we may concede, is that there are configurations realizing these rational skills which contain component skills not all of which are counterfactually robust, i.e., in each case the rational skill is realized partly through empirical component skills. So-called "heuristics," reffered to in the bounded rationality literature (Gigerenzer & Todd, 1999), can be thought of as particular, non-robust configurations that realize rational competences. But this certainly does *not* imply that there exists no other configuration which realizes the rational skill at hand and is just composed of basic analytic skills.

¶48. Turing skills. Given that complex rational skills can be realized through configurations of empirical skills, and assuming that all basic rational skills are analytic, the question turns out to be: Is it true that every complex rational skill can be realized in a sufficiently general, domain neutral way by means of counterfactually robust, i.e., analytic skills?

Let's say that a set of basic skills $T \subseteq B$ is a Turing set if it axiomatizes all rational skills, $\overline{T} = R$. Clearly $\forall T : \overline{T} = R \rightarrow R^\star \subset T$.

The global weak emergence view now boils down to saying there exists an analytic Turing set, $\exists T : \overline{T} = R \wedge T \subset A$. To compare, the local weak emergence hypothesis re rational skill r claims thara is an analytic configuration of r: $\exists \sigma \in \Sigma : \pi(\sigma) = r \wedge \forall s \in \sigma : s \in A$.

The argument from retrieval-and-rule-following configuration (¶46) seeks to identify one such Turing skill set, consisting in the skills to retrieve norms of rationality and to apply them once explicated.

Are there any other Turing sets? This would make the case for global weak emergence more robust.

¶49. Towards empirically testing weak emergence views about the relation between language and thought. Let us suppose that two agents, a and b, possess identical language skills, and assume a masters complex skill $s \in C$. Then:

1. If s is analytic and any configuration σ that functionally realizes s is entirely composed of basic analytic skills ($s \in A \wedge \forall \sigma \forall i : \pi(\sigma) = s \wedge s_i \in \sigma \rightarrow s_i \in A$), then b masters complex skill s.
2. If s is analytic but some configurations σ that functionally realize s ($\pi(\sigma) = s$) contain basic empirical skills ($s \in A \wedge \exists \sigma \exists i : \pi(\sigma) = s \wedge s_i \in \sigma \wedge s_i \notin A$), then a may, but need not master complex skill s.
3. If s is empirical ($s \notin A$), then the more b's basic empirical skills deviate from a's, the more unlikely it is that b masters skill s.

These propositions allow us to deduce, from local (and hence global) weak emergence hypotheses, different empirical consequences which can be be experimentally checked with LLMs. Machine learning experiments with LLMs may hence refute or confirm philosophical theories about the relation between language and thought.

Consider, for example, the local weak emergence view that practical deliberation competence, and specifically the rational skill to balance and weigh conflicting reasons, is a language skill. Suppose we would observe, in machine-learning experiments with systematic competence assessments through trigger condition tests, that (i) every LLM that is fully linguistically competent in English can rationally balance reasons, (ii) LLMs with stronger language skills tend to possess stronger reason balancing skills, and (iii) varying an LLM's empirical skills, e.g. through fine-tuning, tends to have no effect on the LLM's reason balancing skill.—All this empirical evidence would actually confirm the local weak emergence hypothesis.

Put more generally, ontological emergence (¶23) inferentially bridges metaphysical doctrines on the one side and empirical observations we make in using and studying LLMs on the other side.

Notes

[1]https://www.coe.int/en/web/common-european-framework-reference-languages/level-descriptions, retrieved 2025-09-23

References

Arora, S., & Goyal, A. (2023). A theory for emergence of complex skills in language models. Retrieved from https://arxiv.org/abs/2307.15936

Bai, Y., Kadavath, S., Kundu, S., Askell, A., Kernion, J., Jones, A., … Kaplan, J. (2022). *Constitutional AI: Harmlessness from AI feedback*. Retrieved from https://arxiv.org/abs/2212.08073

Bedau, M. A. (1997). Weak emergence. *Philosophical Perspectives*, *11*, 375–399.

Bender, E. M., Gebru, T., McMillan-Major, A., & Mitchell, M. (2021). On the dangers of stochastic parrots: Can language models be too big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. New York, NY, USA: Association for Computing Machinery.

Berti, L., Giorgi, F., & Kasneci, G. (2025). Emergent abilities in large language models: A survey. Retrieved from https://arxiv.org/abs/2503.05788

Betz, G., Richardson, K., & Voigt, C. (2021). *Thinking aloud: Dynamic context generation improves zero-shot reasoning performance of GPT-2*. Retrieved from https://arxiv.org/abs/2103.13033

Betz, G., Voigt, C., & Richardson, K. (2021). Critical thinking for language models. *Proceedings of the 14th International Conference on Computational Semantics (IWCS)*. Association for Computational Linguistics.

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., … Amodei, D. (2020). *Language models are few-shot learners*. Retrieved from https://arxiv.org/abs/2005.14165

Chomsky, N. (1969). Some empirical assumptions in modern philosophy of language. In E. Nagel, S. Morgenbesser, P. Suppes, & M. White (Eds.), *Philosophy, science, and method*. St. Martin's Press.

Da, J., Le Bras, R., Lu, X., Choi, Y., & Bosselut, A. (2021). Analyzing commonsense emergence in few-shot knowledge models. *3rd Conference on Automated Knowledge Base Construction*. London, UK.

Elangovan, A., He, J., & Verspoor, K. M. (2021). Memorization vs. Generalization : Quantifying data leakage in NLP performance evaluation. *ArXiv*, *abs/2102.01818*.

Gigerenzer, G., & Todd, P. M. (1999). *Simple heuristics that make us smart*. New York: Oxford University Press.

Kamoi, R., Zhang, Y., Zhang, N., Han, J., & Zhang, R. (2024). *When can LLMs actually correct their own mistakes? A critical survey of self-correction of LLMs*. Retrieved from https://arxiv.org/abs/2406.01297

Liao, K.-Y., Chang, C.-S., & Hong, Y.-W. P. (2024). A mathematical theory for learning semantic languages by abstract learners. *IEEE Journal on Selected Areas in Communications*, *43*, 2700–2713.

Markov, A. A. (2006). An example of statistical investigation of the text eugene onegin concerning the connection of samples in chains (lecture at the physical-mathematical faculty, royal academy of sciences, st. Petersburg, 23 january 1913). *Science in Context*, *19*, 591–600.

Norvig, P. (2017). On chomsky and the two cultures of statistical learning. In W. Pietsch, J. Wernecke, & M. Ott (Eds.), *Berechenbarkeit der welt? Philosophie und wissenschaft im zeitalter von big data* (pp. 61–83). Wiesbaden: Springer Fachmedien Wiesbaden.

O'Connor, T. (2021). Emergent Properties. In E. N. Zalta (Ed.), *The Stanford encyclopedia of philosophy* (Winter 2021). https://plato.stanford.edu/archives/win2021/entries/properties-emergent/; Metaphysics Research Lab, Stanford University.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., … Lowe, R. J. (2022). Training language models to follow instructions with human feedback. Retrieved from https://arxiv.org/abs/2203.02155

Petroni, F., Rocktäschel, T., Riedel, S., Lewis, P., Bakhtin, A., Wu, Y., & Miller, A. (2019). Language models as knowledge bases? *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2463–2473. Hong Kong, China: Association for Computational Linguistics.

Quine, W. V. O. (1960). *Word and object*. Technology Press of the Massachusetts Institute of Technology.

Quine, W. V. O. (1992). *Pursuit of truth* (2nd ed.). Cambridge, Mass.: Harvard University Press.

Russell, S. J., & Norvig, P. (2016). *Artificial intelligence: A modern approach* (Third edition). Harlow: Pearson.

Ryle, G. (2009/1962). A rational animal. In G. Ryle & J. Tanney (Eds.), *Logic lane. Collected essays 1929-1968* (pp. 428–447). London: Routledge.

Shannon, C. E. (1948). A mathematical theory of communication. *Bell Syst. Tech. J.*, *27*, 623–656.

Sosa, E. (2010). How competence matters in epistemology. *Philosophical Perspectives*, *24*, 465–475.

Wang, X., Antoniades, A., Elazar, Y., Amayuelas, A., Albalak, A., Zhang, K., & Wang, W. Y. (2025). Generalization vs memorization: Tracing language models' capabilities back to pretraining data. *The Thirteenth International Conference on Learning Representations*.

Wei, J., Tay, Y., Bommasani, R., Raffel, C., Zoph, B., Borgeaud, S., … Fedus, W. (2022). Emergent abilities of large language models. *ArXiv*, *abs/2206.07682*. Retrieved from https://api.semanticscholar.org/CorpusID:249674500

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E. H., Xia, F., … Zhou, D. (2022). Chain of thought prompting elicits reasoning in large language models. *ArXiv*, *abs/2201.11903*.

Version

sha: e0996e4dbef85d1f34a5884f806eadd8b7fb5d5ee333d23de48d9a22071c544a

Appendix A. The Tensorization Argument (Arora & Goyal, 2023)

What the Tensorization Argument seeks to accomplish. Given assumptions about the competence of a model on basic skills, we will derive its competence on complex skills. Comnpetence on basic and complex skills are mathematically and apriori tied to each other. More precisely, we will argue that the model's competence on basic skills is k times better than its competence on complex skills which comprise k component basic skills.

Measuring basic skill competence. We're testing the performance of a model $\mathcal{M}$ on basic skills $S = s_1 \dots s_n$ with respect to a large finite **degree-l test stream** consisting of N text pieces $t_1, t_2, \dots t_N$. Each text t_i (i) has been generated with l iid sampled skills, $\{s_{i1} \dots s_{il}\}$, and is augmented with a test-annotation identifying test-tokens; the ability to predict these test-tokens is supposed to track the competence in skills $\{s_{i1} \dots s_{il}\}$.

The performance of the model on a single text piece in the test stream is the **prediction loss**, i.e. the cross-entropy loss with every token that is not marked by the test-annotation being masked.

The competence of the model in a particular skill $\bar{s}_i$ (with $s_i \in S$) is the mean prediction loss averaged over all text pieces generated with s_i.

The overall performance of the model in basic skills averages all skill specific performances.

Conceptualizing complex skills and measuring a model's performance. Given basic skills S, we identify a complex (k-)skill with a k-tuple of basic skills. There are $\binom{n}{k}$ complex k-skills. The performance of the model $\mathcal{M}$ on the set of complex k-skills, $S_k \subset S^k$, can be assessed in close analogy to its performance on basic skills. Thus, we assume that the performance on k-skills can be tested against any large finite **degree-l test stream** consisting of N text pieces $t_1, t_2, \dots t_N$ with the following properties: Each text t_i (i) has been generated with l iid sampled k-**skills**, $\{\mathbf{s}_{i1} \dots \mathbf{s}_{il}\}$, and is augmented with a test-annotation which identifies test tokens to be predicted. The prediction loss on such a degree-l test stream estimates the performance of the model on complex k-skills.

Tensorization of the test stream. Consider a *degree-l test stream* consisting of N text pieces $t_1, t_2, \dots t_N$ suitable to assess the model performance on basic skills, i.e., each text piece is generated with l basic skills. Let us randomly shuffle and re-number so as to obtain the following stream:

$$(1) \qquad \boxed{t_1 | s_{11}, s_{12}, \dots, s_{1l}}\,\boxed{t_2 | s_{21}, s_{22}, \dots, s_{2l}} \dots \boxed{t_j | s_{j1}, s_{j2}, \dots, s_{jl}} \dots$$

Note that we're also plotting the set skills associated with each text piece. Let us next partition this stream by concatenating any k consecutive text pieces, leaving test annotations intact:

$$\boxed{t_1 + t_2 + ... + t_k | s_{11}, s_{12}, ..., s_{1l}, s_{21}, s_{22}, ..., s_{2l}, ..., s_{k1}, s_{k2}, ..., s_{kl}}$$
$$(2) \quad \boxed{t_{k+1} + ... + t_{2k} | s_{(k+1)1}, s_{(k+1)2}, ..., s_{(k+1)l}, ..., s_{(2k)1}, s_{(2k)2}, ..., s_{(2k)l}}$$
$$\boxed{t_{2k+1} + ... + t_{3k} | s_{(2k+1)1}, s_{(2k+1)2}, ..., s_{(3k)l}} \ ...$$

Let's call this the k-concatenated degree-l test stream.

This k-concatenated degree-l test stream is eo ipso a degree-kl test stream. As we're using masked cross-entropy loss to calculate prediction loss and assess model competence, prediction loss adds up, an the average prediction loss on a text piece in the k-concatenated degree-l test stream is **k-times** the average prediction loss on a text piece in the original degree-l test stream.

Next, we can shuffle and randomly split up the kl skills associated with each text piece in the k-concatenated degree-l test stream (Equation 2) into l partitions. Each partition is a k-tuple of skills that can be identified with a complex k-skill $\mathbf{s} \in S_k$. We assume that text generation from skills is compositional in the following sense: Each concatenated text piece can be conceived of as being generated by, and hence being associated with, the corresponding complex skills. This gives us:

$$\boxed{t_1 + t_2 + ... + t_k | \mathbf{s}_{11}, ..., \mathbf{s}_{1l}}$$
$$(3) \quad \boxed{t_{k+1} + ... + t_{2k} | \mathbf{s}_{21}, ..., \mathbf{s}_{2l}}$$
$$\boxed{t_{2k+1} + ... + t_{3k} | \mathbf{s}_{31}, ..., \mathbf{s}_{3l}} \ ...$$

Yet this is nothing but a iid sampled **degree-l test stream** for **k-skills**, suitable to test the performance of a model on k-skills. We've argued that the prediction loss on this test-stream is k-times the prediction loss of the original test stream used for assessing the model competence on basic skills. Model competence, measured by prediction loss, therefore scales with the number of component skills when moving from basic to complex skills.

Appendix B. The 1-Skill-Learner Model (Liao, Chang, & Hong, 2024)

What is the 2-Skill-Learner model supposed to show? The 1-Skill-Learner model developed by Liao et al. (2024) establishes the possibility of curricular emergent skills and helps to quantify the extent of curricular emergence.

The degree-c training stream. We assume that there are n basic skills S and N training texts T. Each training text $t \in T$ is associated with $k(t)$ uniformly sampled skills, where $k(\cdot)$ are independent Poisson random variables with mean c. We call a specific skill graph sampled from a corresponding random bipartite graph a **degree-c training stream**.

Learning, Understanding, Curricular Emergence. Training on *degree-c training stream* proceeds by repeatedly presenting all texts in the stream to the model. This may allow the model to learn novel skills. Training stops as soon as the model has learned all skills.

A model *understands* a text t if it has learned all skills associated with t.

We shall say that there is curricular emergent understanding if, after training, the model *understands* a text associated with skill set $\mathbf{s}$ without having seen a text associated with this particular skill tuple during training. This can also be cast as the model having acquired a complex skill (tuple of basic skills) it has not been trained on.

The 1-Skill Learner. A system is a 1-skill learner (with latency l) iff it learns a skill s from one text t just in case s is the only *novel* skill associated with t, i.e. the only skill the system has not learned before (or, if this has occurred l times for skill s during training – like Liao et al. (2024), we shall just consider the case $l = 1$).

Emergent understanding is almost trivially possible in this model, e.g.: The system learns *all skills* by been presented with texts that are associated with at most 2 skills, it has hence *not* seen any texts that instantiate triple skills, yet is understanding such texts, too.

Beyond mere possibilities, the formal analysis of the abstract 1-Skill-Learner model by Liao et al. (2024) allows to quantify the extent of emergent understanding and curricular emergent complex skills. We assume that out-of-sample test texts are chosen similarly to the training texts.

Figure 1 demonstrates that the error of a 1-skill learner drops sharply once the training corpus is scaled beyond a certain, relatively small threshold. To see why this establishes curricular emergence, consider $c = 4$. According to Figure 1, the model learns all skills at threshold $R \approx 1.5$. Let's assume there are $n = 20$ different skills. That means that a degree-4 training stream of size $N = 30$ will typically enable a model to learn all skills by training on this corpus for consecutive epochs. With $c = 4$, we may expect that all of these 30 texts in the training corpus contain between 1 to (at most) 10 skills. However, there are more than a million ($2^{20} - 1$) complex skills that may in principle be associated with texts; half of these combine more than 10 skills and are very unlikely to be present in the tiny training corpus.

Emergent understanding his hence not only possible, but massive.

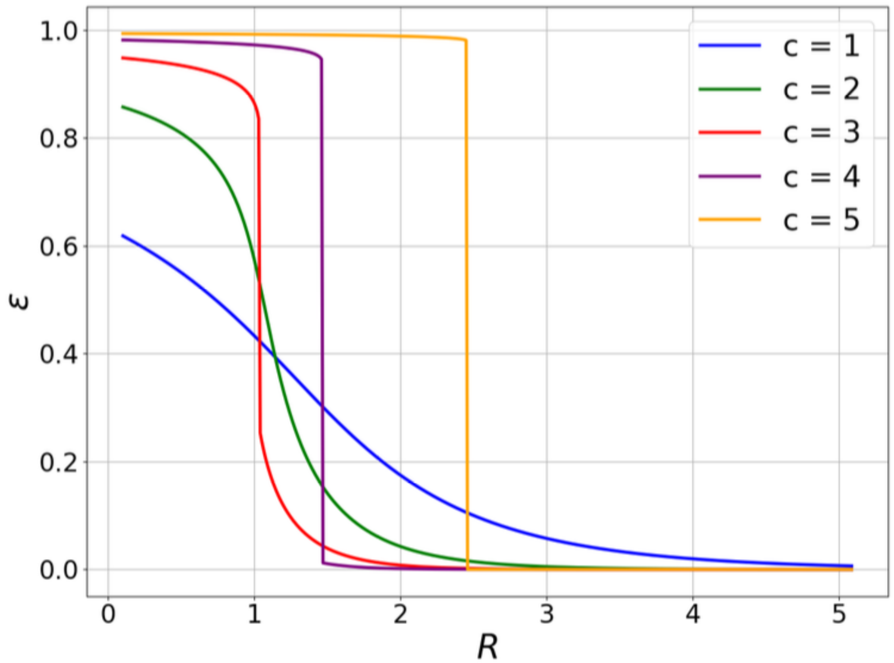

FIGURE 1. Expected error ϵ of 1-skill learners on out-of-sample texts and as function of the ratio of training text corpus size to the number of basic skills, i.e. $R = N/n$. Lines show results for different Poisson distributions of text degrees, namely with texts being associated with, on average, $c = 1 \ldots 5$ skills. Source: Liao et al. (2024).